MIL-C-915B(NAVY)
AMENDMENT - 3
20 October 1967
SUPERSEDING 1/
Amendment - 2
13 March 1967

MILITARY SPECIFICATION

CABLE, CORD AND WIRE, ELECTRICAL (SHIPBOARD USE)

This amendment forms a part of Military Specification MIL-C-915(NAVY), 23 April 1965, and has been concurred in by the Marine Corps.

Page 1, paragraph 1.2.1.1: Opposite Type MDY, delete "steel tape, armored".

Pages 2 and 3, paragraph 2.1: Add the following specifications:

SPECIFICATIONS

FEDERAL
"PP-V-205 - Veneer, Paper-Overlaid, Container Grade.
"QQ-W-461 - Wire, Steel, Carbon (Round, Bare and Coated).

MILITARY
"MIL-L-17331 - Lubricating Oil, Steam Turbine, Non-Corrosive.
"MIL-C-17621 - Cable, Telephone, Electric, Submarine.
"MIL-I-45208 - Inspection System Requirements.

Delete the listing for MIL-M-20693 and substitute:

"MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid."

Page 3, paragraph 2.2: Delete reference to "Circular NBS 431".

Page 4, paragraph 3.1.1, line 1: After "types" insert "CVSF".

Page 4, paragraph 3.2, lines 1 and 2: Add "TFU-6".

Page 6, paragraph 3.9.2, item (d): Add "(including issue letter)".

Page 13, paragraph 3.11.4.2, last sentence: After "be" insert "substantially".

Page 23, paragraph 3.12.18.5, line 2: Delete "1800" and substitute "1400".

1/CHANGES FROM PREVIOUS ISSUE.- THE OUTSIDE MARGINS OF THIS DOCUMENT ARE MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.,) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 6145

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Page 24, paragraph 3.12.19.1.6, listing, second column: Delete and substitute:

"3 turns on mandrel 3 x diameter of cable.
1 turn on mandrel 4 x diameter of cable.
180-degree turn on mandrel 5 x diameter of cable."

Pages 26 and 27, paragraph 3.12, test requirements, group C, line 2: Delete "(see 3.12.2.1.4)" and substitute "(see 3.12.2.1.3)".

Pages 29 and 30, paragraph 3.15, Construction, Sixth: Delete "(see 3.2.18)" and substitute "(see 3.12.18)".

Page 31, paragraph 3.16, test requirements, Group C, line 2: Delete "(see 3.12.2.1.4)" and substitute "(see 3.12.2.1.3)".

Pages 32 and 33, paragraph 3.17, test requirements, Group A, last line: Under "Requirements" column, add "1.0".

Pages 32 and 33, paragraph 3.17, test requirements, Group C, line 2: Delete "(see 3.12.2.1.4)" and substitute "(see 3.12.2.1.3)".

Pages 35 and 36, paragraph 3.19: Under Construction, First, line 1: Delete "49 by 0.023" and substitute "7 by 7 by 0.023"; test requirements, group A, insulation resistance, delete "at 25° C," and substitute "at 15° C."; under Capacitance, line 5, column 2, delete "40 ± 2.5" and substitute "45 max. (no min.)"; Hydrostatic, line 2, delete "1000 p.s.i." and substitute "500 p.s.i."; Pressure cycling, line 2, column 2, delete "1000" and substitute "500"; Group C, delete the first item in both columns.

Page 36, table XVIII: Title, delete "DDS" and substitute "DSS"; last line, column 4" Delete ".025" and substitute ".020".

Pages 42 and 43, paragraph 3.24: Under Construction, First, line 1: Delete "49 by 0.023" and substitute "7 by 7 by 0.023".

Pages 47 and 48, paragraph 3.28: Under Construction, Second, delete "(see 3.12.2)" and substitute "(see 3.12.3)"; under test requiements, Group C, line 2, column 1, delete "(see 3.12.5.1.4)" and substitute "(see 3.12.2.1.3)"; and under group C, line 3, column 2, delete "80" and substitute "90".

Pages 50 and 51, paragraph 3.30, test requirements, group C, line 2: Delete "(see 3.12.2.1.4)" and substitute "(see 3.12.2.1.3)".

Pages 54 and 55, table XXXI, column 2: Item 7, delete "259(37 x 7)" and substitute "500(259)"; item 8, delete "427(61 x 7)" and substitute "650(427)"; and item 30, delete "427(61 x 7)" and substitute "600(427)".

Pages 54 and 55, table XXXI, column 8, item 6: Delete".0141" and substitute ".0453".

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Pages 60 and 61, paragraph 3.37, group A: Under conductor resistance delete reference to "Size 2-1/2 (26)" and "4.35".

Page 65, paragraph 3.41: Delete and substitute:

"3.41 Type TPU-6 cable.- 1/

CONSTRUCTION

First - A conductor comprising 19 strands of 0.0095 inch steel wire, steel number 1010, finish 5, medium, class 1 in accordance with QQ-W-461.
Second - A wall of synthetic resin insulation (see 3.12.2) having a minimum wall thickness (see 3.8.3) of 0.025 inch.
Third - Two conductors shall be twisted together with a 2-1/4-inch lay to form a twisted pair.
Fourth - Six twisted pairs shall be cabled together. The length of lay shall be suitable to meet the bending test on the completed cable. Color coding (see 3.10.7.2).
Fifth - An approved binder over the assembled pairs.
Sixth - A polychloroprene sheath (see 3.12.17). Reinforcement (see 3.12.12.3) shall be applied under the sheath, or approximately near the middle of the sheath thickness so as not to show as an overall surface marking. The minimum sheath thickness shall be not less than 0.125 inch.

Type TPU-6 replaces cable covered by MIL-C-17621.

Test Requirements

Group A	Requirements
Visual and dimensional (see 4.7)	
Dielectric strength (see 4.8.19)	
Volts, r.m.s., minimum	
Conductor to conductor	2000
Conductor to water	1000
Insulation resistance (see 4.8.25)	
Megohms per 1000 feet at 25° C.	100
Watertightness (see Appendix I)	
Cubic inches in 6 hours	1

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Group B	Requirements
Insulation (see 3.12.2)	
Tensile - p.s.i., minimum	1800
Elongation - percent, minimum	100
Sheath (see 3.12.17)	
Tensile-p.s.i., minimum	1800
Elongation - percent, minimum	300
Set - inch, maximum	3/8
Bending (see 4.8.14)	
Bending cycles at -30° C.	500
Breaking strength (see 4.8.28) pounds, minimum	1100

Group C	Requirements
Insulation (see 3.12.2.1.3)	
Elongation - percent of original value, minimum	80
Sheath (3.12.17.4 and 3.12.17.5)	
Tensile - percent of original value, minimum	65
Elongation - percent of original value, minimum	65
Conductor	
Physical requirements (prior to stranding)	
Steel number 1010, finish 5, medium, class 1 of QQ-W-461.	

Table XLIX - Type TPU-6 cable.

Type	Number of pairs	Diameters	
		Overall insulated conductor	Overall
		Minimum	Maximum
TPU-6 1/	6	0.105	1,000

1/ The nominal length or unit ordering length for type TPU-6 is 1000 feet. No length under 250 feet will be accepted.

"3.42 Workmanship. - Workmanship shall be first class in every respect and uniform throughout each lot of cable, cord, or wire offered for delivery."

Page 66, paragraph 4.1.1: Delete and substitute:

"4.1.1 Inspection system.- The supplier shall have an inspection system which meets the requirements specified in MIL-I-45208. The inspection system shall include the establishment and maintenance of in-process inspections to be performed at appropriate locations within the supplier's plant for the purpose of assuring continuous control of the manufacture of components and assemblies."

Page 66, paragraph 4.2, line 1: After "types" insert "CVSF".

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Page 66: Add as paragraphs 4.2.2, 4.2.2.1, and 4.2.3:

"4.2.2 Periodic qualification re-evaluation.- Re-evaluation may be requested by Naval Ship Engineering Center at any time after the date of the letter of notification of the qualification of the product. Materials for evaluation shall be from current production.

"4.2.2.1 Re-evaluation procedure.- When re-evaluation is requested, it shall be the responsibility of the qualified supplier to furnish the Government with the data necessary to establish the continued conformity of the product to all qualification requirements. These data should preferably be complete test results of a specimen from current production, tested against all the requirements of the specification. At the discretion of the Government, test records from current production may be accepted for the re-evaluation to the extent that they are available. Specimens from current production need to be subjected only to the tests for which no production test records are available. If a failure occurs, no cable represented by the specimen shall be offered for delivery until the cause for failure has been determined and corrective measures taken. Final acceptance of product is contingent upon the product's meeting all the qualifying requirements of the specification.

"4.2.3 Qualification specimens.- At least one specimen of cable for which individual type or group qualification is sought shall be submitted for qualification inspection. The cable type, size, and length of specimens required for group qualification shall be in accordance with the following listing:

Group	Qualifying type and size	Length of specimen (feet	Cables types in group
1	DHFR-4	200	SHFR, DHFR, THFR
2	TTHFWA-10	200	TTHFWA, PBTM, TSP-- all sizes
5	DSS-3	200	DSS, TSS, FSS, 7SS--all sizes
6	MCSF-4	200	MCSF
7	FCSF-66	200	FCSF
8	SDU-500	200	SDU-- all sizes
9	MHOF-30	200	MHOF-- all sizes
10	TTOP-10	200	TTOP-- all sizes
11	MCOS-6	200	MCOS-- all sizes
12	MMOP-5	200	MMOP
13	THOF-9	200	DHOF, THOF, FHOF-- sizes 9 and smaller
14	THOF-150	200	DHOF, THOF, FHOF -- sizes 14 and larger. CVSF and T3P--all sizes
15	TTRS-8	200	TTRS, TTRSA-- all sizes

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The groups are based on similar characteristics and requirements. The Government may extend approval of any one cable type and size to cover any or all types in the same group. The Government may require performance of any or all qualification inspections on additional types in the same group prior to such extension of approval."

Page 69, table XL, column 1, last item: Delete "181" and substitute "131".

Page 69, paragraph 4.4.6.3.5 Delete and substitute:

"4.4.6.3.5 Action in case of rejection.- Any completed cable which does not comply with specifications and cannot be made to comply and is not acceptable to the Government shall be recorded with the Government inspector. The written report shall be furnished within 10 days after rejection, and shall show the cable type, size, lengths, and serial numbers of the marker tape recorded from each end of each length."

Page 69, paragraphs 4.4.6.4 through 4.4.6.5.4: Delete.

Pages 70 and 71, paragraph 4.5.2: Add the following test:

Test	Paragraph reference
Flexing endurance	4.8 14"

Page 71, paragraph 4.5.3: Delete the following test:

Test	Paragraph reference
"Flexing endurance	4.8.14"

Pages 71 and 72, paragraph 4.5.4: Delete and substitute:

"4.5.4 Verification and comparison inspection.- To insure that manufacturers continue to supply cables which conform to specification requirements, verification and comparison inspections, in addition to inspections normally conducted by a Government inspector, shall be performed on all cable requiring Government source inspection at the manufacturer's plant.

"4.5.4.1 Verification inspection.- Verification inspection shall be conducted at the manufacturer's plant under the surveillance of a Government representative. The inspection shall consist of the applicable tests of groups A, B,

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and C as required for quality conformance inspection. The sample used and test results obtained from it may be used as a part of the quality conformance inspection for that particular lot.

"4.5.4.1.1 Sampling procedure for verification inspection.- One sample shall be selected from every fifth log, as a minimum, of each type and size of cable for which qualification is required.

"4.5.4.2 Comparison inspection.- Comparison inspection shall be conducted outside the manufacturer's plant, at a Government designated laboratory. The following tests shall be performed as applicable:

Test	Paragraph reference
Insulation and sheath	
Synthetic resin, synthetic rubber, butyl rubber, polythene, and polychloroprene	4.8.1
Metal armor	4.8.7
Drip	4.8.8
Cold bend	4.8.9
Flammability	4.8.10
Pressure cycling	4.8.31.2
Flexing endurance	4.8.14
Impact resistance	4.8.15
Abrasion resistance	4.8.16
Hot oil test	4.8.17
Induced voltage	4.8.29
Dielectric	4.8.19
Flame retardance	4.8.26
Capacitance and characteristic impedance	4.8.27
Breaking strength	4.8.28
Heat shock	4.8.30
Hydrostatic	4.8.31
Thermoplastic flow	4.8.11
Watertightness	Appendix I

"4.5.4.2.1 Sampling procedure for comparison inspection.- One sample shall be selected by a Government representative for comparison inspection from the first lot of each type and size of cable produced by a manufacturer after his product has been qualified. Thenceforth, one sample of the type and size required for qualification shall be selected every 4 months, or as near as possible

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to every 4 months. If the specific type and size is not available for sampling at the proper time, a sample from the lot of the nearest type and size that falls within a qualifying group should be used.

"4.5.4.2.2 Length of specimens.- The length of specimens required for submittal shall be as follows:

Cable groups 1, 2, 7, 10, and 12. - 15 feet
Cable groups 6, 9, 13, and 14. - 20 feet
Cable groups 8 and 11. - 25 feet
Cable groups 5 and 15. - 50 feet

"4.5.4.2.3. Marking of specimens.- All test specimens shall be marked with the following information:

(a) Name of manufacturer.
(b) Cable designation (type and size)
(c) Manufacturing lot number or order number either of the command or agency concerned or the shipbuilder (if submitted for delivery).
(d) Footage represented by the specimen.
(e) The coil or reel number from which sample came.
(f) Qualification designation.

"4.5.4.2.4 Packaging of specimens for shipment.- To facilitate handling, specimens may be shipped in boxes or wrapped coils.

"4.5.4.3 Action in case of failure.- If any specimen fails one or more of the specified tests, the lot represented by that specimen shall be held in abeyance. The manufacturer shall have the option of resubmitting the lot, but only after the lot has been reevaluated, the cause of failure determined, and corrective action has been taken. If the manufacturer resubmits the lot, the specimen representing the lot shall be subjected to the test or tests under which failure occurred, plus any other tests which may be deemed necessary to assure compliance to this specification. If the manufacturer chooses not to resubmit the lot, or if the lot is resubmitted and a failure is found, the lot shall be rejected. Once Government inspection reveals a failure in a specific type and size of cable, all new lots of the same type and size shall be submitted for Government inspection until two successive lots pass all inspections. After two successive lots pass Government inspection, normal testing shall be resumed.

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"4.5.4.4 Action in case of rejection.- Any completed cable which has been rejected and is not acceptable to the Government shall be recorded with the Government inspector. The written report shall be furnished within 10 days after rejection and shall show the cable type, size, lengths, and serial numbers of the marker tape recorded from each end of each length."

Page 71, paragraph 4.6: Delete and substitute:

"4.6 Reports.- One copy of laboratory reports covering comparison inspection shall be forwarded to the Naval Ship Engineering Center."

Page 76, paragraph 4.8.7.1, line: After "application", delete "of" and substitute "to".

Page 79, paragraph 4.8.12.1, line 3: Delete "MHFF" and substitute "MHOF".

Page 82, paragraph 4.8.17.2, lines 1 and 2: Delete "Navy Symbol No. 3100 of Pamphlet NBS 431" and substitute "type 2190 TEP of MIL-L-17331".

Page 83, paragraph 4.8.19.6.5, last line: Delete "oil" and substitute "foil".

Page 88, paragraph 4.8.29.3.3, line 1: Delete and substitute:

"4.8.29.3.3 Calculations.- Mutual inductance in henries shall be less than 0.05 microhenries.

Page 89, paragraph 4.8.31.1, line 3: Delete "815-F-1197101" and substitute "815-F-1197030".

Page 89, paragraph 4.8.31.2, line 3: Delete "815-F-1197101" and substitute "815-F-1197030".

Page 90, paragraph 5.1.1.2.2. line 4: Between "plywood" and "shall" insert "or weather-resistant veneer, paper-overlaid conforming to PPP-V-205."

Page 96, Index of Cable Types: Item 18, delete "MHFF" and substitute "MHOF".

Type	Paragraph number	Page
"TPU-6	3.41	65"

Page 100, INDEX, page 100; After "Tolerances, dimensional; Add the following listing:

"Verification inspection 4.5.4".

Pages 105 through 107, Appendix II: Delete.

Other interest:
MC

Preparing activity:
Navy - SH
(Project 6145- N020)

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